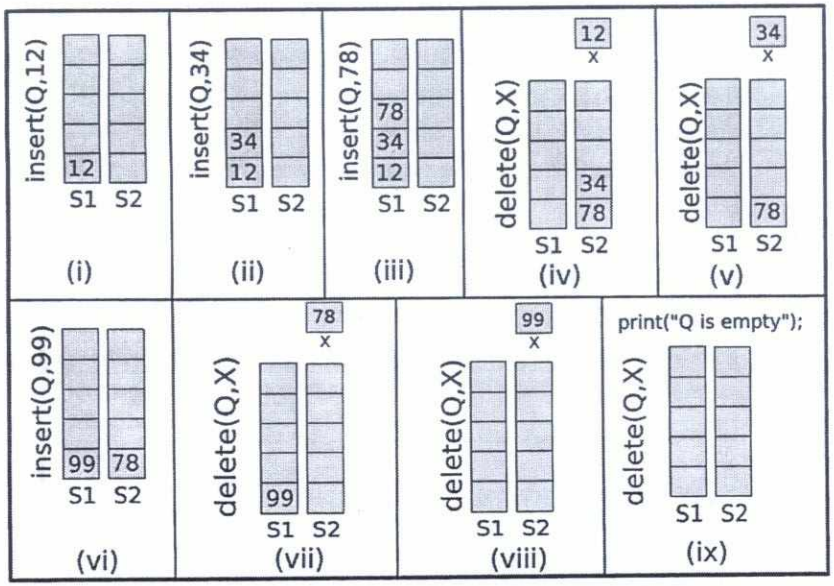


Figure 4 Figure for Problem 5

[4]

Thapar Institute of Engineering and Technology, Patiala
Electrical and Instrumentation Engineering (EIED)

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

UCS540/UBM605

Data Structures and Algorithms

MID-SEMESTER TEST (MST)

BE-**EE/EIC/BME VI Semester**, 2022-23

March 18, 2023 Saturday, 3.30 PM

Duration: 2 Hrs Max. Marks: **50**

Name of Faculty : Dr. Ashish Gupta and Dr. Mukesh Dalal

General Instructions: All questions are compulsory. Write concise and precise answers. Draw neat, clean, and labeled diagrams wherever required. Bloom's Taxonomy notations: L-❶: Remembering, L-❷: Understanding, L-❸: Applying, L-❹: Analyzing, L-❺: Evaluating, L-❻: Creating.

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1. Consider the instance of Tower of Hanoi problem shown in Figure 1, where the objective is to achieve the goal configuration from the initial configuration.

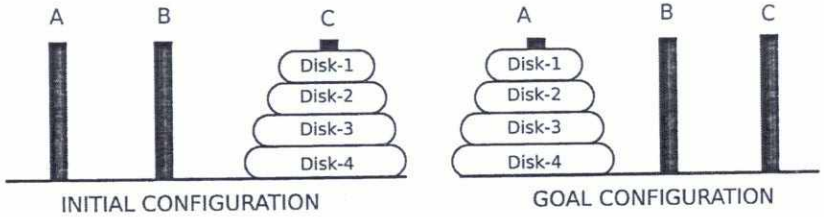


Figure 1: Figure for Problem 1

   (a) Write a generic recursive algorithm to accomplish the task.

   (b) Find the sequence of moves required to complete the task for n=4 by drawing the tree of calls / recursion tree using your algorithm.

   (c) Comment on the running time of the algorithm.

   (L-❸, 4+4+2 marks)

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2. (a) Let ↑ represents exponentiation and assuming normal operator precedences, convert each of the following infix expressions into its prefix and postfix equivalent forms:

   (i) A+B+C*D*E↑F

   (ii) A↑ B*C+D-E/F/(H-G) **(L-❸, (0.75×2)×2 marks)**

   (b) Write a procedure/algorithm to convert an infix expression into a postfix expression and demonstrate its working on A+B+C*D*E↑F. **(L-❸, 4 marks)**

   (c) How can a postfix expression be computed using a stack machine? Illustrate the steps of your procedure/algorithm on the output of Part (b) of Problem 2 with A=B=C=2 and D=E=F=3. **(L-❸, 3 marks)**

3. (a) Consider the instance of the string matching problem given in Figure 2.

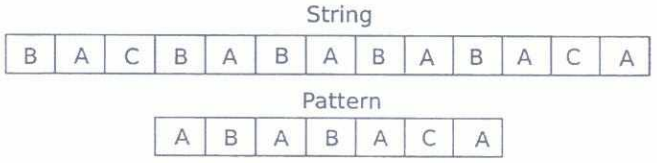


Figure 2: Figure for Problem 3

   (i) Compute the LPS array for the Pattern given in Figure 2.

   (ii) Illustrate the working of KMP algorithm by applying it on the given instance of string matching problem.

(L-❸, 2+3 marks)

   (b) Sort the following list using Insertion sort. Show each pass of the sort and discuss its run time complexity

   78, 23, 56, 33, 59, 99, 12, 10, 90, 75 **(L-❶, 3+2 marks)**

4. (a) Write the algorithm for Binary search and discuss its limitations and advantages over linear search. **(L-❶, 2+2 marks)**

[2]

   (b) In some language that supports negative indexes, a "C" array declaration **Data[9][6][6]** can be written as **Data[-1:7, -5:0, 7:12]**. Let the base address of the array be 500 and each element occupies 4 bytes of memory. Compute the address of **Data[3][-2][10]** if the elements are stored in a row major order? **(L-❹, 3 marks)**

   (c) An array representation of a list is shown in Figure 3. Each element of the list is mapped to an array position using the formula: **location(i) = (6+i) % 10**, where % is the modulo division operation (produces the remainder of an integer division). In other words, i^{th} list element is stored at (6+i)%10 index in the array as can be seen in Figure 3.

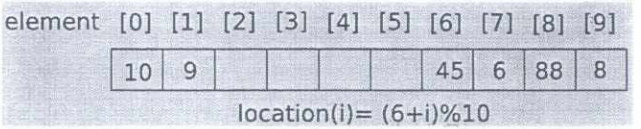


Figure 3: Mapping [45,6,88,8,10,9] into a 1-D array.

Write an algorithm to insert an element at any valid position in the list using the provided array representation.

(L-❺, 3 marks)

5. Consider a queue Q that is implemented using two stacks S1 and S2. A call to insert(Q, item) inserts an element at the rear of the queue and a call to delete(Q, X) deletes the front element and assigns it to a variable X. Figure 4 shows the consecutive snapshots ((i) to (ix)) of such an implementation of Q when elements are inserted and deleted in the order shown starting with an empty queue. Observe the procedures for insertion and deletion carefully to design an algorithm for

   (a) insert(Q, item) using push and pop operations only.

   (b) delete(Q, X) using push and pop operations only.

   (c) Trace your algorithms to demonstrate their correctness.

[3] **(L-❻, 2+5+3 marks)**